

Superstore Sales & Profit Analysis Report

1. Project Overview

This project analyzes the sales, profit, discount, quantity, shipping, and regional performance of a global superstore dataset using Python and Microsoft Power BI.

The main objective is to identify profitable products, loss-making areas, customer behavior, and business opportunities.

2. Data Cleaning & Schema Evolution

Data cleaning is a critical step in ensuring data integrity and usability. The initial dataset (superstore.csv) contained redundant columns, non-standard naming schemas, and unparsed temporal variables. The transformed dataset (clean_superstore.csv) retains all \$51,290\$ transactional rows while streamlining the structural schema from 27 attributes down to 20.

Uncleaned Column (superstore.csv)	Cleaned Column (clean_superstore.csv)	Transformation Applied / Rationale
Customer.ID	Customer_id	Renamed dot-notation to standard snake_case for SQL/Python compatibility.
Order.Priority	Order_priority	Renamed to snake_case.
Product.Name	Product_name	Renamed to snake_case.
Sub.Category	Sub_category	Renamed to snake_case.
Ship.Mode	Ship_mode	Renamed to snake_case.
Shipping.Cost	Shipping_cost	Renamed to snake_case.
Order.Date & Ship.Date	<i>Dropped</i>	Extracted temporal features (Year, weeknum, Month) and dropped raw text timestamps.

Uncleaned Column (superstore.csv)	Cleaned Column (clean_superstore.csv)	Transformation Applied / Rationale
None	Month	Feature Engineering: Extracted integer month values \$[1 - 12]\$\$ from order dates for seasonality analysis.
记录数 (Record Count)	Dropped	Removed redundant counter variable (always equals \$1\$).
Customer.Name	Dropped	Anonymized dataset for compliance and confidentiality.
Order.ID, Product.ID, Row.ID	Dropped	Dropped unique system hashes/identifiers to optimize memory footprint during aggregation.
Market2	Dropped	Removed redundant geographical classification.

3. Exploratory Data Analysis (EDA) & Core Insights

The cleaned dataset covers \$51,290\$ records with the following high-level financial metrics:

- Total Sales Generated: \$12,642,905
- Total Profit Realized: \$1,467,457.29
- Total Units Sold (Quantity): \$178,312\$ items

A. Temporal Trends & Growth Metrics

The superstore experienced massive financial expansion over the 4-year tenure. Sales and profits scaled steadily year-over-year:

- **2011:** Sales: \$2,259,511\$ | Profit: \$248,940.81
- **2012:** Sales: \$2,677,493\$ | Profit: \$307,415.28
- **2013:** Sales: \$3,405,860\$ | Profit: \$406,935.23
- **2014:** Sales: \$4,300,041\$ | Profit: \$504,165.97

B. Product Performance Analysis

A breakdown by category shows that while Technology brings in the highest absolute revenue and profit, Office Supplies maintains a remarkably high profit-to-sales margin.

1. Technology: Sales = \$4,744,691 | Profit = \$663,778.73
 2. Furniture: Sales = \$4,110,884 | Profit = \$285,204.72
 3. Office Supplies: Sales = \$3,787,330 | Profit = \$518,473.83
- Top 5 Sub-Categories by Sales: Phones \$1,706,874 , Copiers \$1,509,439 , Chairs \$1,501,682, Bookcases \$1,466,559 and Storage \$1,127,124.

C. Geographical

- Top 5 Most Profitable Regions: Central \$311,403.98), North \$194,597.95, North Asia \$165,578.42, South \$140,355.77, and Central Asia \$132,480.19.

4.Key Business Insights

These are the Insights that has been found by Power BI Dashboard and Pandas.

Sales & Profit Insights

1. **Phones generated the highest sales** and were among the top profit-generating sub-categories, showing strong customer demand and profitability.
2. **Copiers produced the highest profit** while maintaining relatively low discount levels, making them one of the most efficient product categories.
3. The company should primarily focus on:
 - Phones
 - Copiers
 - Bookcases
 - Appliances
 - Chairs
 - Storage

These categories contribute significantly to overall revenue and profit.

4. **Higher sales generally lead to higher profit**, but the relationship is only moderate. Some high-sales products still produce low profit because of heavy discounts or high operational costs.
5. **Technology category recorded the highest overall sales**, especially during November and December.
6. **Office Supplies category showed stable profitability**, indicating consistent customer demand.

Discount Analysis

7. **Discounts above 25% often resulted in losses**, reducing overall profitability.
8. **Discounts above 40% created major losses**, indicating over-discounting negatively impacts the business.
9. **Discount strategies did not significantly increase sales performance**, suggesting aggressive discounting is not an effective long-term growth strategy.
10. **Increasing discounts reduces profit margins**, proving that excessive discounting harms profitability.

11. Products with lower discounts, especially Copiers and Phones, maintained healthier profit margins.

Monthly & Seasonal Trends

12. **November and December recorded the highest sales and profits**, mainly because of year-end and festive season purchasing behavior.
13. **February showed comparatively lower sales performance**, making it one of the weaker months for business activity.
14. Sales and profit consistently increased during the last quarter (Q4), indicating strong seasonal demand.

Regional & City Insights

15. Some cities generated high sales but low profit due to taxation and discount policies.
16. **Aba city showed relatively high tax impact**, affecting net profitability.
17. APAC region had the highest average sales among all regions.
18. Canada showed high average profit despite lower quantity sold, indicating efficient sales performance.

Shipping Insights

19. **Standard Class shipping handled the majority of sales and profit**, making it the most preferred shipping method.
20. Same Day shipping contributed the least to total sales and profit.

Quantity Insights

21. Higher quantity orders generally improved sales performance.
22. However, large quantity sales with heavy discounts often reduced profitability.

Business Recommendations

- Reduce discounts above 25%.
- Avoid excessive discounting on already profitable products.
- Increase inventory and marketing for high-performing categories like Phones and Copiers.
- Focus more on Q4 campaigns because November and December generate maximum revenue.
- Improve profitability in low-performing cities by reviewing tax and pricing strategies.
- Promote profitable low-discount products instead of relying heavily on discounts.

Dashboard Summary

The dashboard successfully provides:

- Sales forecasting
- Profit tracking
- Category-wise analysis
- Shipping analysis
- Regional comparison
- Monthly sales trends

- Discount impact analysis

using Microsoft Power BI visualizations.

Short Conclusion

The analysis shows that business growth is driven mainly by Technology products, especially Phones and Copiers. While discounts increase customer purchases slightly, excessive discounting significantly reduces profitability. Seasonal demand during November and December creates the highest revenue opportunities. The company should focus on profitable products, controlled discount strategies, and optimized regional operations to improve long-term business performance.